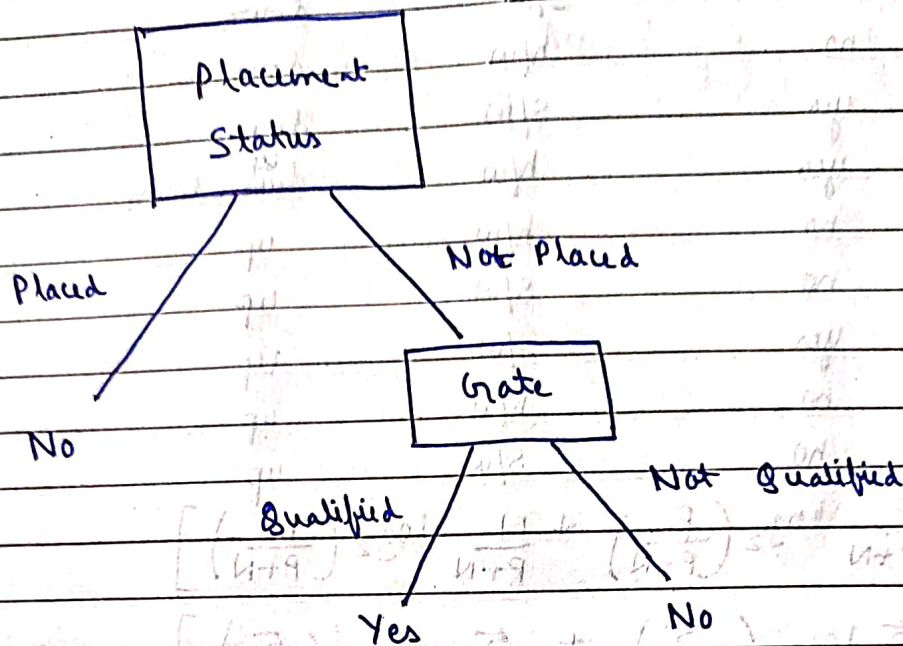


* Decision Tree and Random Forest



How to select ordering of features?

A decision tree chooses the feature that best separates the data at each step. The feature which results in the cleanest split.

Random Forest is a group of many decision trees that work together and vote to make a final decision.

Random Forest combines many decision trees to give a more accurate and stable prediction.

Tree → one opinion

Forest → many opinion → better decision.

Age	Competition	Type	Profit
old	yes	s/w	down
old	no	s/w	down
old	no	h/w	down
mid	yes	s/w	down
mid	yes	h/w	down down
mid	no	h/w	up
mid	no	s/w	up
new	yes	s/w	up
new	no	h/w	up
new	no	s/w	up

$$I.G = - \left[\frac{P}{P+N} \log_2 \left(\frac{P}{P+N} \right) + \frac{N}{P+N} \log_2 \left(\frac{N}{P+N} \right) \right]$$

$$I.G = - \left[\frac{5}{10} \log_2 \left(\frac{5}{10} \right) + \frac{5}{10} \log_2 \left(\frac{5}{10} \right) \right]$$

$$= - \left[0.5 \log_2 2^{-1} + 0.5 \log_2 2^{-1} \right]$$

$$= - \left[-0.5 - 0.5 \right]$$

$$= - \left[-1 \right]$$

$$I.G = 1$$

Age

$$\text{Gain} = I.G - E(A)$$

$$\text{Entropy} = I.G \times \text{Probability}$$

	Down	Up
old	3	0
mid	2	2
new	0	3

$$I(\text{old}) = - \left[\frac{3}{3} \log_2 \left(\frac{3}{3} \right) + \frac{0}{3} \log_2 \left(\frac{0}{3} \right) \right] \times \frac{3}{10} = 0$$

$$I_{mid} = - \left[\frac{2}{4} \log_2 \left(\frac{2}{4} \right) + \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \right] \times \frac{4}{10} = 1 \times \frac{4}{10} = 0.4$$

$$I_{new} = - \left[\frac{1}{3} \log_2 \left(\frac{1}{3} \right) + \frac{3}{3} \log_2 \left(\frac{3}{3} \right) \right] \times \frac{3}{10} = 0 \times \frac{3}{10} = 0$$

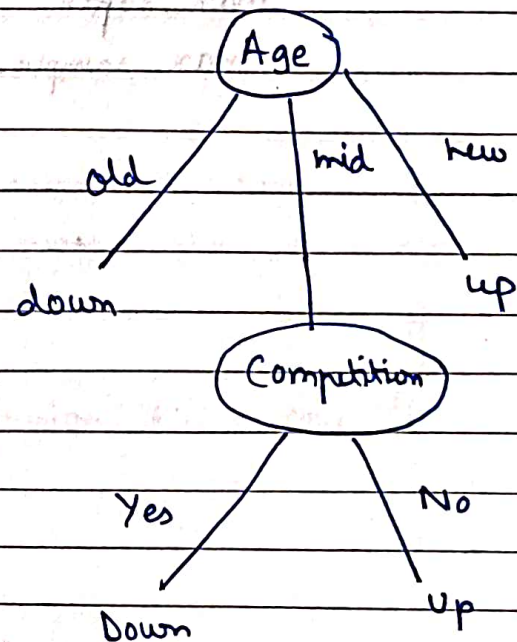
0.4

$$\begin{aligned} \text{Gain} &= I.G - E(A) \\ &= 1 - 0.4 \\ &= 0.6 \end{aligned}$$

$$\text{Gain (Age)} \rightarrow 0.60 \quad (\text{Root})$$

$$\text{Gain (Competition)} \rightarrow 0.124$$

$$\text{Gain (Type)} \rightarrow 0$$



* When we build the tree in depth overfitting problem comes up

* Pruning technique:

If the tree overfits then we need to use pruning technique.

Age	Gender	Bought
29	Male	Yes
34	Male	Yes
13	Female	Yes
27	Female	No
10	Female	NO

- Growing the tree beyond a certain level of complexity leads to overfitting.
- A model is not going to add any value. Hence we can cut it.
- This process of trimming trees is called Pruning.

Pruning parameters:-

max-leaf-nodes

max-depth

max-sample-leaf

$$(A) \rightarrow P \cdot I = \text{node}$$

$$P \cdot O - 1 =$$

$$2 \cdot O =$$

(node)

$$0 \cdot 0 = (\text{node}) \text{ node}$$

$$P \cdot O - 1 = (\text{node}) \text{ node}$$

$$0 = (\text{node}) \text{ node}$$